AWS and EC2 Instances

Region

1. Region is nothing but the separate geographic area like (Mumbai/ Ohio/ Singapore/) where the AWS has its infrastructure.
2. Each region is completely independent of other regions.
3. Regions are used for low latency, data compliance, and disaster recovery.
4. **Example:** Mumbai Region = ap-south-1
5. There are up to **39** regions in AWS

Availability Zone

1. An Availability Zone is a group of one or more data centres inside a region.
2. Each AZ is isolated from others (separate power, network, etc.).
3. Used for high availability and fault tolerance. (if one AZ fails, others still work).
4. A region has multiple AZs (like Mumbai has multiple AZs).
5. There are also **120+** availability zones in AWS and one region contains up to 3 availability zones
6. **Example** (Mumbai Region):
   - ap-south-1a
   - ap-south-1b
   - ap-south-1c
7. only North Virginia having the 6 availability zones.

Example:

- Region = Mumbai City
- Availability Zone = Buildings inside that city

CLOUDBLITZ

"If Mumbai (Region) has a power failure in one building (AZ),
other buildings (AZs) still keep running — so applications don't go down."

STEPS TO CREATE A UBUNTU INSTANCE

1. name ----------> ubuntu
2. ami ----------> ubuntu
3. instance type ---> t3.micro
4. key pair ----------> ubuntu
5. vpc ----------> default
6. sg/firewall -------> default
7. storage ----------> default

8. create a instance

Important Components

AMI (Amazon Machine Image)

1. AMI is a pre-configured template that contains OS, applications, and settings.
2. It helps to launch EC2 instances quickly without installing everything manually.
3. You can create custom AMIs from your own instance for reuse and backup.

Instance Type

1. Instance type defines the hardware configuration (CPU, RAM, storage, network).
2. Different types are designed for different workloads (General, Compute, Memory, etc.).
3. You can change instance type based on performance needs (scaling).

Key Pair

1. Key pair only support the key-based authentication not password-based authentication.
2. Key Pair is used for secure login to EC2 instances (SSH for Linux, RDP for Windows).
3. It consists of Public Key (stored in AWS) and Private Key (kept by user).
4. Private key must be kept safe, because it cannot be recovered if lost.

CLOUDBLITZ

Security Group

1. Security Group acts as a virtual firewall controlling inbound as well as outbound traffic.
2. You can allow specific ports (like 22 for SSH, 80 for HTTP).
3. It is stateful, meaning return traffic is automatically allowed.
4. In security group, it only allows the traffic not denied the traffic.

EBS (Elastic Block Store)

1. EBS provides persistent block storage for EC2 instances.
2. Data is not lost when instance stops (unlike temporary storage).

INSTANCE STATE

1. **Start** ----------> Moves instance from stopped → running
2. **Stop** ----------> Moves instance from running → stopped
3. **Terminate** ----------> Permanently deletes the instance
4. **Hibernate** ----------> Sleep mode (saves RAM state)
